

7. For vessels in Cyclic service or when WRC limits are exceeded, the analysis should be performed with an industry accepted and Contractor/Company approved Finite Element Method. Allowable stress limits shall be in accordance with Part 5 of ASME Section VIII, Division 2.
8. For Vessels which are designed for cyclic service, Nozzles shall be designed for a total number of minimum 7000 full temperature cycles over expected life times.
9. For tangential nozzles, the local load analysis on shell/head and nozzle neck shall be based on FEA.
10. If the application of the loadings, specified in this appendix, to a fully reinforced nozzle assembly, result in an increase in the shell or head thickness over that required for pressure and support loadings, then SUPPLIER is to advise the maximum loadings that can be applied to that reinforced nozzle assembly, without having to increase the shell or head thickness over that required for pressure and support loadings.
11. Contractor shall use actual nozzle forces & moments, in case nozzle sizes are larger than NPS 24.

Table A2-2 - Nozzle Allowable Forces and Moments

Pipe size (NPS)	ASME Class	P (N)	VL, VC (N)	FR (N)	MT (Nm)	ML, MC (Nm)	MR (Nm)
2	150	1000	1225	2000	350	250	495
	300	1000	1225	2000	350	250	495
	600	1485	1820	2975	470	335	665
	900	1485	1820	2975	470	335	665
	1500	1800	2205	3600	530	375	750
	2500	1800	2205	3600	530	375	750
3	150	1510	1850	3020	825	585	1170
	300	1510	1850	3020	825	585	1170
	600	2045	2500	4085	1070	755	1510
	900	2855	3500	5710	1380	975	1955
	1500	3705	4535	7405	1645	1160	2325
	2500	3705	4535	7405	1645	1160	2325
4	150	2150	2635	4300	1540	1090	2180
	300	2150	2635	4300	1540	1090	2180
	600	2985	3655	5940	2050	1450	2900
	900	3785	4640	7575	2485	1760	3515
	1500	5450	6720	10975	3260	2305	4610
	2500	5450	6720	10975	3260	2305	4610
6	150	3780	4630	7560	4075	2880	5765
	300	4600	5630	9200	4860	3440	6880
	600	5695	6975	11390	5865	4145	8295
	900	7250	8880	14505	7185	5080	10160
	1500	10595	12975	21185	9605	6795	13585
	2500	10740	13150	21475	9700	6860	13720
8	150	5690	6970	11380	7615	5385	10770
	300	6060	7425	12125	8075	5710	11420
	600	7100	8700	14205	9325	6595	12190
	900	12100	14820	24200	14785	10455	20910
	1500	16005	19600	32005	18415	13020	26045
	2500	17865	21880	35725	19950	14110	28215

Pipe size (NPS)	ASME Class	P (N)	VL, VC (N)	FR (N)	MT (Nm)	ML, MC (Nm)	MR (Nm)
10	150	8070	9880	16135	12755	9020	18040
	300	10910	13360	21820	16820	11895	23780
	600	12840	15730	25680	19460	13760	27520
	900	17795	21795	35595	25755	18210	36425
	1500	22920	28075	45845	31555	22315	44630
	2500	27150	33250	54300	35800	25315	50625
12	150	9880	12100	19755	17520	12390	24780
	300	10665	13065	21335	18830	13315	26630
	600	17665	21635	35330	29840	21100	42200
	900	25010	30630	50020	40250	28460	56925
	1500	32930	40330	65860	50160	35470	70940
	2500	37630	46085	75255	55395	39170	78340
14	150	10875	13320	21795	19870	14050	28100
	300	12640	15485	25285	22895	16200	32375
	600	21150	25905	42300	36635	25905	51810
	900	30050	36805	60095	49580	35060	70115
	1500	41830	51235	83665	64465	45585	91170
	2500	61185	74940	122370	83345	58935	117870
16	150	12470	15275	24940	24340	17215	34425
	300	16495	20200	32985	31700	22415	44830
	600	27225	33345	54445	50125	35445	70890
	900	38345	46965	76690	67405	47660	95325
	1500	52085	63795	104170	86180	60940	121880
	2500	74145	90815	148290	110400	78065	156130
18	150	14065	17230	28130	28665	20270	40535
	300	20855	25545	41715	41630	29435	58870
	600	34060	41715	68120	65210	46110	92220
	900	48650	59585	97295	88770	62770	125540
	1500	65855	80660	131710	113180	80030	160065
	2500	93195	114140	186385	144465	102155	204310
20	150	15050	18435	30105	35175	24875	49745
	300	21810	26715	43620	50020	35370	70745
	600	37005	45320	74005	81250	57455	114910
	900	52505	64305	105010	110055	77820	155645
	1500	67175	82275	134345	134465	95085	190165
	2500	99125	121405	198250	178055	125905	251810
22	150	15630	19140	31260	41790	29550	59105
	300	22110	27080	44215	58120	41100	82200
	600	38865	47600	77730	97650	69050	138100
	900	54795	67110	109585	131605	93060	186120
	1500	74895	91840	149970	169585	119915	239830
	2500	107155	131245	214310	218390	154425	308855
24	150	16670	20420	33340	50955	36030	72065
	300	22755	27870	45510	68475	48420	96840
	600	39400	48255	78800	113445	80220	160440
	900	57040	69865	114085	156400	110590	221180
	1500	76260	92405	152520	197660	139770	279535
	2500	108605	133390	217810	254550	179995	359990